The Palaeomagnetism of the Stormberg Lavas of South Africa

1: Evidence for a Genuine Reversal of the Earth's Field in Triassic–Jurassic Times

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Summary

The lowermost lavas of the Stormberg basalts are reversely magnetized and the uppermost lavas normally magnetized with a transition zone in between where the declination and inclination change progressively.

Several different self-reversal mechanisms are reviewed in the light of a variety of field and experimental evidence, all of which support the hypothesis of a genuine reversal of the main geomagnetic field during Triassic–Jurassic times when the lavas were extruded.

1. Introduction

A palaeomagnetic study has been made of the Stormberg basaltic lavas of Triassic–Jurassic age occurring in Basutoland. The lava succession was sampled systematically in two areas 80 miles apart. The first area sampled was at Sani Pass on the eastern side of the Drakensberg (29° 18′ E, 29° 36′ S), where the lavas attain a maximum thickness of about 4 000 ft. The second set of samples was collected in the Maseru area (27° 48′ E, 29° 24′ S) where the lavas are approximately 3 000 ft thick. The sites are marked 1 and 2 on Figure 1.

A total of 151 cores with an average length of 3 ft were drilled, using the coring rig designed by Graham & Keiller (1960). 132 samples (i.e. cores) were collected from the lavas, 6 samples from the underlying aeolian Cave Sandstone where it had been baked by the lavas, 3 samples from the underlying unbaked Cave Sandstone, one sample from a thin sandstone lens with a baked upper contact about 1 100 ft up in the lava succession, one sample from a sandstone zenolith 180 ft above the base of the lavas and 8 samples from three dolerite dykes and the adjacent lavas cut by them. The dolerite dykes are genetically associated with the lavas and are thought to be feeders for this vast fissure type of eruption (du Toit 1954).

The declination, D, and the inclination, I, are plotted against the elevation in feet above the base of the lava sequence in Figure 2(a). In general, 10 specimens were measured from each core. The 99 per cent confidence limits (α) were calculated by Fisher's method (Fisher 1953) and are also shown on the diagram.

It is evident that the lowermost lavas are magnetized in the opposite direction to the present field, i.e. are reversely magnetized, whereas the uppermost lavas are magnetized in a direction close to that of the present field, i.e. are normally magnetized. There is a transition zone in which both D and I change progressively.

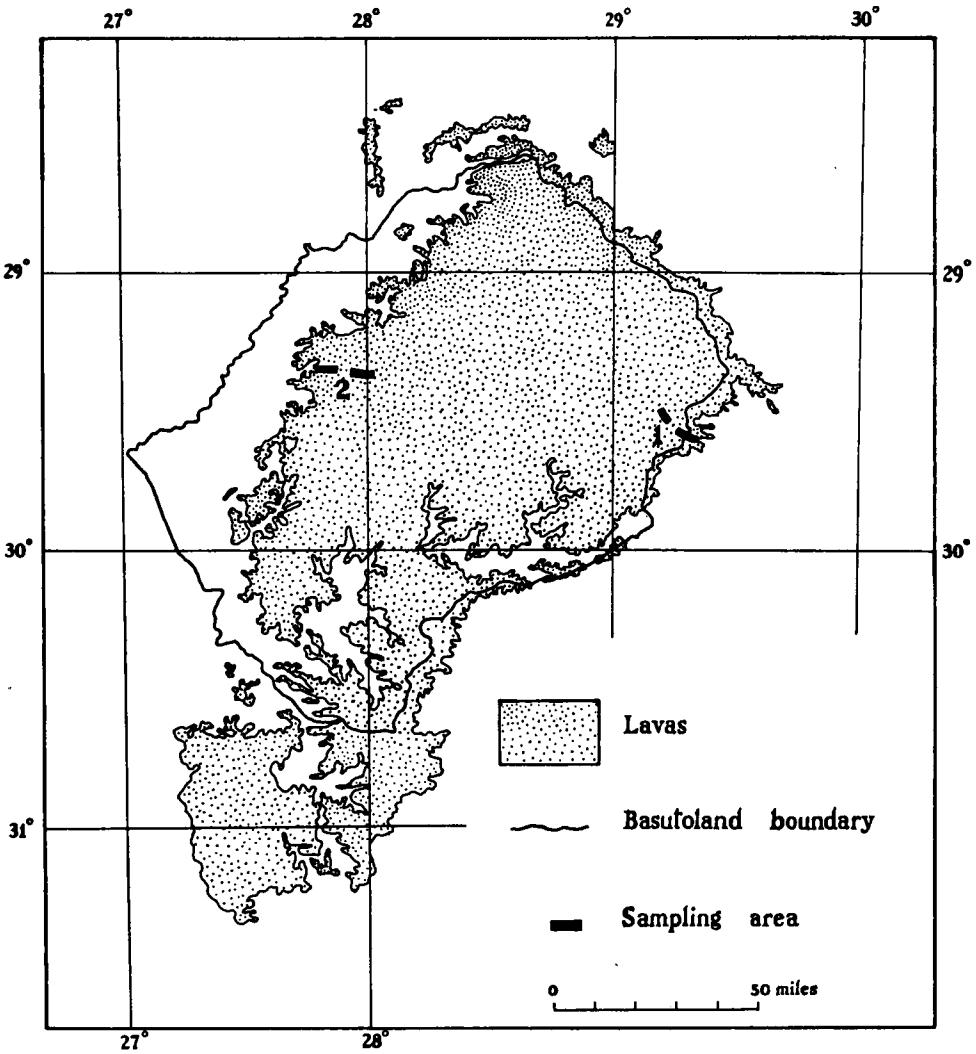


FIG. 1.—Sketch map of Basutoland showing extent of basalts.

Figure 2(b) shows the mean D and I for 4 specimens from each core after a.c. demagnetization in a field of 219 Oe (500 c/s). The points marked by crosses are from specimens which did not show tight grouping after demagnetization. It is thought that these specimens had been magnetized by lighting currents. This aspect of the investigation together with an analysis of the behaviour of the field intensity during the transition period will be reported in a separate paper.

Comparing Figure 2(b) with Figure 2(a) it can be seen that there is less scatter from core to core and there is a remarkable correlation between the results for the two areas, especially in the transition zone.

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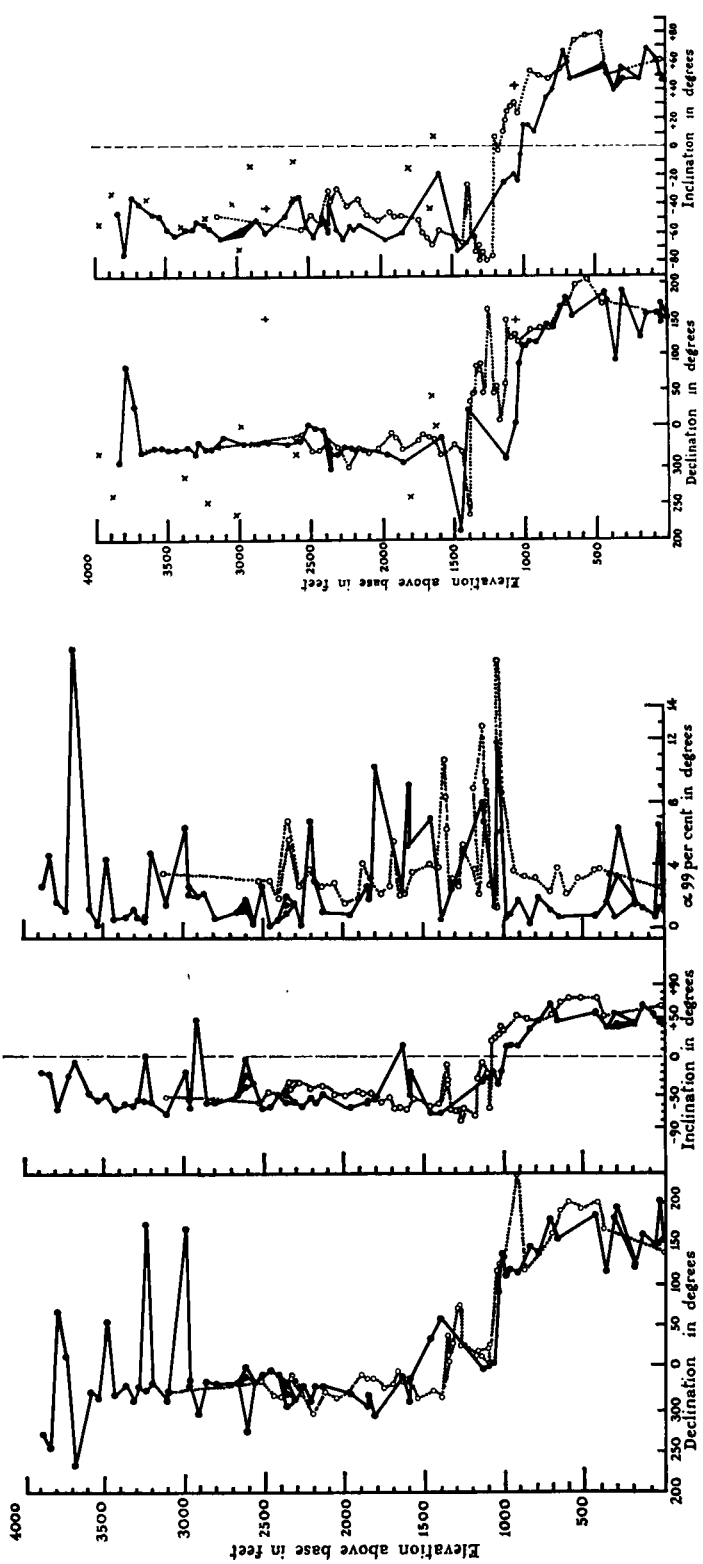


FIG. 2.—(a) *D*, *I* and α 99 per cent against elevation above base of lava succession.

(b) *D* and *I* against elevation above base of lava succession after a.c. demagnetization at 219 Oe.

× specimens from Sani Pass which did not show tight grouping.

+ specimens from Maseru area which did not show tight grouping

—●— Sani Pass area

·····O····· Maseru area

D measured as direction of N pole from N clockwise.

I negative when N pole above horizon.

Points joined by more than one line constitute cores from same flow.

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This paper discusses the measurements which were made in order to establish that the reverse magnetization in the lower lavas arose as a result of a reversal of the Earth's field.

2. The self-reversal hypotheses

Néel (1951, 1955) and others have suggested several mechanisms whereby rocks cooling in the Earth's field can acquire a magnetization in the opposite sense to that field. These are summarized below.

(i) In certain substances which possess spinel structure an exchange interaction between the sublattices of the two constituents can take place. The two sublattices have spontaneous magnetizations opposite to one another, and if the temperature variations of the spontaneous magnetizations differ, the substances may show spontaneous self-reversal (Néel 1951). Magnetite is a spinel and a common constituent in lavas and it is possible therefore that it could behave in this way. However, no natural specimen showing this behaviour has so far been found.

(ii) Hospers (1953) has suggested a modification of Néel's hypothesis. If in a two-component magnetic mineral the spontaneous magnetization of the component B is greater than that of the component A and in the opposite sense, then initially the nett magnetization will be in the direction of B. If now the magnetically strong ions in B are replaced by feebly magnetic ions, the direction of magnetization may change to that of the A component, i.e. will be in the opposite direction. Again, such a process has not been observed.

(iii) There may be two components A and B with different Curie points in the rock. Suppose that the Curie point of A is the greater then A becomes magnetized in the direction of the external field. The magnetic field at the time B passes through its Curie point will be the resultant of the external field and the demagnetizing field of A. This resultant may be negative, so that when B passes through its Curie point it may be magnetized in a field opposite to that of the external field (Néel 1951). If the magnetization of the B component is greater than that of the A component at room temperature, then the rock will show reverse magnetization. The coupling between the two constituents may be either magnetostatic or by exchange interaction across the boundaries. It has been suggested that the reversal in the Haruna dacite is due to an exchange interaction. In this case the mineral assemblage responsible for the self-reversal belongs to the ilmenite–hematite solid solution series (Uyeda 1958, p. 57).

(iv) Even if in mechanism (iii) the magnetization of the A component is the greater so that initially the rock is magnetized in the direction of the external field, chemical changes may occur as a result of which the magnetization of the A component decreases and in time that of the B component will become dominant. Graham's oxidation process (Graham 1953) comes under this heading.

(v) Néel (1955) and Verhoogen (1956) have suggested that a mineral may undergo self-reversal when cations migrate to preferred sites in the crystal lattice. This has been described as migration from a disordered state to an ordered state. If the cooling is rapid as in laboratory experiments, the cations may be frozen in the disordered state for very long periods. Verhoogen estimates that the ordering process as a result of which self-reversal occurs, may take 10^5–10^6 years for impure magnetites.

Summarizing the self-reversal hypotheses, (i) involves a self-reversing magnetic mineral, (iii) takes into account the interaction of two different ferromagnetic ore

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minerals, (ii) and (iv) involve chemical changes so that normally magnetized rocks differ chemically from reversely magnetized ones, and (v) involves a time-dependent self-reversal.

3. The *D* and *I* curves

Figure 2(a) showed the variation of *D* and *I* with height above the base of the lava succession. Since the uppermost lavas are magnetized in a direction close to that of the present field it is possible that the measured magnetization of these lavas is due to viscous magnetization acquired comparatively recently. In order to investigate this point a number of specimens were demagnetized in an a.c. field of 500 c/s. The demagnetizing equipment has been described by Graham (1961). Figure 3 shows the demagnetization curves for a number of specimens from the normal and reversed zones. It was noted that the direction of magnetization of some specimens settled down to a steady value at about 100 Oe, while that of the majority of specimens remains constant to about 350 Oe. Four specimens from each core were then demagnetized in fields of 219 Oe and the mean *D* and *I* for these specimens are shown in Figure 2(b). As has been remarked already the demagnetized specimens show rather less scatter between cores. The upper lavas still show normal magnetization and the lower lavas reverse magnetization. We are now left with the possibility that the difference in the directions of magnetization is due to a difference in the magnetic minerals present in the lower and upper lavas as a result of which the lower group show self-reversal whereas the upper do not.

In order to investigate this possibility four specimens from each core were heated to 650 °C and cooled in the Earth's field. The specimens were placed with known orientation in a brass box sealed by means of a copper gasketed lid held in position with stainless steel screws. The box was evacuated to a pressure of about 2 cm Hg for several minutes and then filled with nitrogen. Nitrogen was allowed to pass through the box thoughout the heating, which lasted 1½ h, and the cooling. The directions of magnetization after the cooling were found without exception to be in the Earth's field within the experimental error.

As a further check, an experiment was carried out in which a representative set of samples was cooled over a period of 4 days. In this case the specimens were contained in a vessel which was evacuated to 1 mmHg, filled with nitrogen and then sealed. The vessel was then placed in a kiln at a brickworks. The maximum temperature in the vicinity of the box was 790 °C as measured by means of pyrometric cones. Once again, the measured directions after cooling were in the direction of the Earth's field, within experimental error. The results of this experiment considered in conjunction with the X-ray analyses exclude all self-reversal mechanisms except those such as (v) which are time-dependent.

The differences in the magnetic components of the upper and lower lavas which are required by the self-reversal hypotheses should be revealed by ore microscope and X-ray studies.

X-ray diffractometer analyses were made of unheated and laboratory heated specimens of lavas from the normal and reversed zones. Silicon metal was used as the internal standard. The unit cell dimensions of the magnetic oxides found from these analyses are given in Table 1. The magnetite present is of fairly uniform composition in all the cases investigated, the average unit cell being 8·41 Å. This unit cell dimension corresponds, using the curve of Vincent & others (1957,

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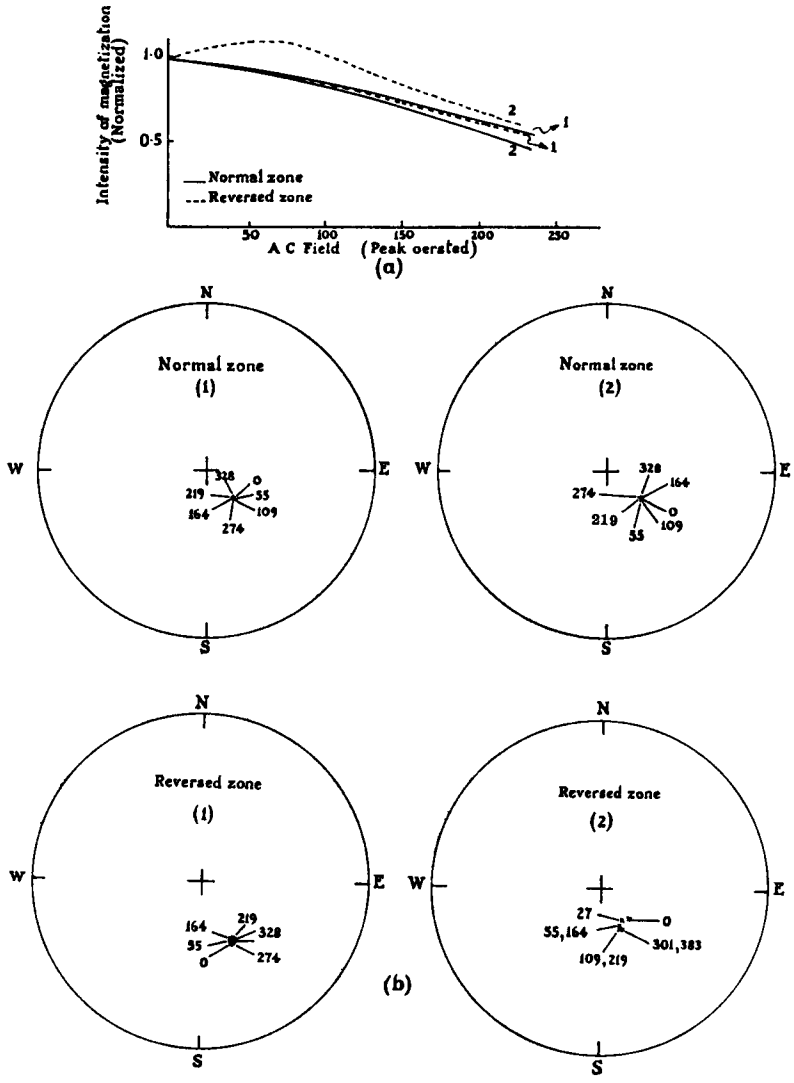


FIG. 3.—(a) A.C. demagnetization curves of typical normal and reversed zone specimens.

(b) Directions of magnetization of these specimens at increasing a.c. fields.

Lower hemisphere plots
⊙ south poles down
× north poles down
164—value of a.c. field in peak oersteds.

p. 632) to a magnetite with 15–25 molecular per cent ülvospinel (Fe_2TiO_4) in solid solution. The ilmenite is in all cases except two, a pure ilmenite with cell dimensions corresponding very closely to those given by Basta and quoted by Nicholls (1955, p. 137). In one of the exceptions, a normally magnetized, extremely fine-grained unbaked lava, the ilmenite contains about 66 molecular per cent hematite in solid solution. This estimate was made on the basis of Vincent's curve (1957, p. 640). The second exception is also a normally magnetized unbaked lava in which

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the composition of the rhombohedral phase corresponds to an ilmenite with approximately 50 molecular per cent hematite in solid solution.

There was no evidence from the X-ray examination that the composition of the titano-magnetite phases had been changed by the heating in the laboratory.

A number of specimens were examined under the ore microscope. The results of the examination are given in Table 2. In all cases examined under the ore microscope magnetite with exsolution of ilmenite along the (111) planes of the

Table 1

Lava type	Cell dimensions of magnetite phase in Å	Cell dimensions of ilmenite phase (referred to hexagonal cell) in Å	
		a_H	c_H
Nh	8·41	5·085	14·04
N	8·41	5·043	13·83
N	8·43	5·061	13·87
N	8·43	5·082	14·00
N	8·43	5·083	14·03
N	8·41	5·080	14·03
T	8·40	5·083	14·03
T	8·42	5·084	14·03
T	8·40	5·082	14·03
T	8·44	5·079	14·01
Rh	8·41	5·087	14·04
R	8·42	5·082	14·04
R	8·40	5·077	14·00
R	8·44	5·078	13·99
R	8·42	5·084	14·03
R	8·42	5·084	14·04
RbN	8·41	5·084	14·04

Nh = Normal zone lava heated to 650 °C in N_2.
N = Normal zone lava.
T = Transition zone lava.
Rh = Reversed zone lava heated to 650 °C in N_2.
R = Reversed zone lava.
RbN = Reversed zone lava baked by normally magnetized dyke and remagnetized normally.

host were recognized. The amount of the exsolved ilmenite varies from specimen to specimen. In addition, unorientated discrete mostly elongated grains of ilmenite occur. No systematic difference in the relative abundances of ilmenite grains and magnetite–ilmenite intergrowths between the normal and reversed lavas was found. It should be noted, however, that comparatively little of the material was studied.

It is thought that the occurrence of the 66 mol per cent Fe_2O_3 in solid solution in one of the ilmenites from the normal zone is to be ascribed to chilling soon after the ilmenite–hematite phase commenced to exsolve from the magnetite host, for this specimen showed very fine-grained ore minerals when examined under the ore microscope. The other exception was not examined under the ore microscope.

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Table 2

Lava type	Silicates Vol. %	Magnetite–ilmenite exsolution Vol. %	Exsolution lamellae of ilmenite in magnetite Vol. %	Discrete ilmenite grains Vol. %	Average size of magnetite–ilmenite grains (mm.)	Average size of discrete ilmenite grains (mm.)	Occurrence of magnetite–ilmenite grains	Occurrence of discrete ilmenite grains
Normal zone lava (unheated)	91·7	0·1	0·04	8·2	0·08	0·05	Mostly equi-dimensional	Mostly as needle-like crystals.
Normal zone lava (heated to 650°C in N_2)	89·6	8·3	0·7	2·1	0·3	0·2	do.	Mostly as irregular crystals.
Reversed zone lava (unheated)	93·8	1·2	0·2	5·0	0·4	0·2	do.	As needle-like crystals, as very small (0·05 mm) irregular crystals and seldom as long rectangular crystals.
Reversed zone lava (heated to 650 °C in N_2)	84·6	6·0	1·8	9·4	1·0	1·5	do.	As needle-like crystals and as irregular grains.
Reversed zone lava baked normal by normal dyke	93·7	4·0	1·2	2·3	0·5	0·5	As equi-dimensional and irregular grains	Mostly as irregular grains.
Reversed zone lava baked normal by dyke (heated to 650°C in N_2)	89·5	6·0	1·2	4·5	1·3	1·0	As equi-dimensional and rectangular grains	As needle-like grains and irregular grains.
Transition zone lava (unheated)	94·8	2·5	0·7	2·7	0·2	0·2	do.	do.

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There is thus no reason to believe that any chemical difference exists between the normal and reversed lavas, but rather that chemical differences are attributable to critical cooling conditions during the cooling of the magma such as would be expected at the edges of flows, or in very thin flows. The fact that there is no systematic variation in the thickness of the flows throughout the succession (Stockley 1947, p. 52) leads the authors to believe that the hypothesis of chemical difference correlatable with the direction of magnetization is untenable. This conclusion would serve to exclude all the self-reversal hypotheses including (v) since (v) requires either that there should be a difference in the mineral content of the upper and lower lavas, or that there should be a systematic difference in cooling conditions due to difference in flow thickness.

It is now fairly well established that certain minerals belonging to the ilmenite–hematite solid solution series can give rise to self-reversed magnetization. The titano–magnetite is presumably irrelevant to the problem (Nagata & Uyeda 1959, p. 890).

The ilmenite–hematite series $x\mathrm{FeTiO_3}.(1-x)\mathrm{Fe_2O_3}$ is paramagnetic when $1 > x > 0{\cdot}75$, ferromagnetic when $0{\cdot}75 > x > 0{\cdot}5$ and parasitically ferromagnetic when $0{\cdot}5 > x > 0$. Self-reversal only takes place in the range $0{\cdot}67 > x > 0{\cdot}45$ due to exchange interaction between ferromagnetic (low Curie point) and a parasitically ferromagnetic (high Curie point) phases (Uyeda 1958). For $x \fallingdotseq 0{\cdot}1$ there is a tendency for reversal of thermo-remanent magnetization (T.R.M.) and Uyeda (1958, p. 100) is of the opinion that the reversely magnetized rocks of the Adirondacks may be caused by such an "imperfect" reversal mechanism.

As seen from Table 1, the member of the ilmenite–hematite series present in the reversed zone and the normal zone is a pure ilmenite and is therefore paramagnetic. For the remaining two lavas the ilmenite–hematite members correspond to $x = 0{\cdot}33$ which according to Uyeda's figure (1958, p. 98) corresponds to a member with a normal T.R.M. and to $x \doteq 0{\cdot}5$ which is capable of producing self-reversal. The composition of the exsolving phase is critically dependent on temperature control during cooling. If exsolution is allowed to occur in a titano-magnetite solid solution series ($\mathrm{Fe_3O_4}$—$\mathrm{FeTiO_3}$) the spinel and rhombohedral phases change in composition until they approach pure $\mathrm{Fe_3O_4}$ and $\mathrm{FeTiO_3}$ respectively (Vincent & others 1957). Only where the ores suffer rapid chilling in a restricted temperature range would a $x\mathrm{FeTiO_3}(1-x)\mathrm{Fe_2O_3}$ solid solution with $x \fallingdotseq 0{\cdot}6$ or $x \fallingdotseq 0{\cdot}1$ be expected to form. In this connection it may be noted that the Haruna dacite is a highly vesicular acid lava pointing to critical cooling conditions. These critical conditions are unlikely in igneous rocks. It is for this reason that very few rocks have been described in which ferri-ilmenite lamellae have been identified in the magnetite (Vincent & others 1957, p. 654). Moreover it would be expected that one would find self-reversal only where such special conditions could have prevailed, e.g. near the edge of flows or in thin flows. In the present study many such specimens were collected but no case was found in which the directions of magnetization in the chilled zone differed from those elsewhere in the flow.

Summing up this discussion it is clear that the ilmenite member present in in the typical reversed and normal lavas is almost pure paramagnetic ilmenite and so far as present laboratory investigations show, does not give rise to self-reversal.

4. The directions of magnetization in the transition zone

It will be noted from Figures 2(b) and 4(a) that the directions of magnetization in the transition zone differ very considerably from those of both the normal and

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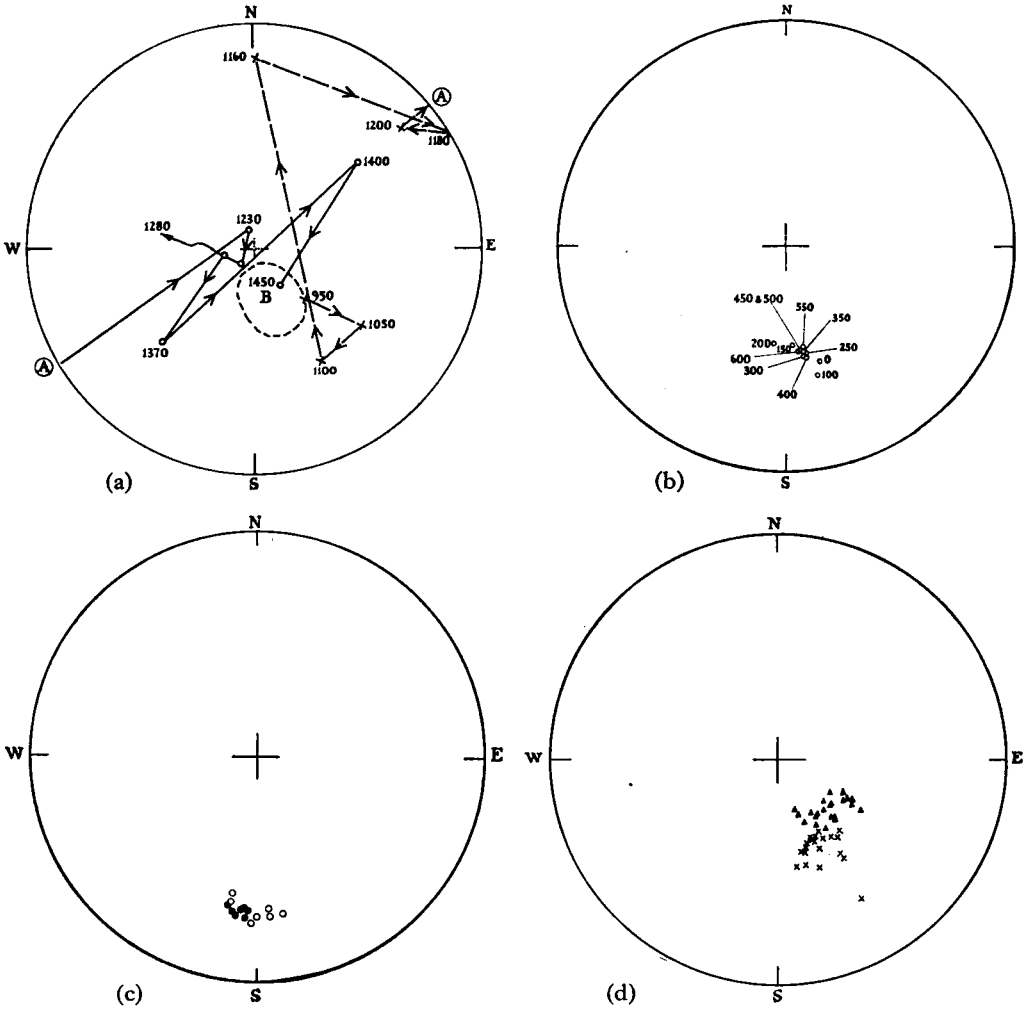


FIG. 4.—(a) Changes in the directions of magnetization in the transition zone in the Maseru area.

Lower hemisphere stereographic plot.
○ south poles down × north poles down
The numbers show height in feet above base of lavas
Note: (i) Ⓐ follows on Ⓐ as directions change from reversed to normal.
(ii) the area B is not a circle of confidence but denotes the region occupied by typical normal and reversed zone lavas.

(b) Directions of magnetization of transition zone lava during thermal demagnetization.
Lower hemisphere stereographic plot.
○ south poles down
The numbers show temperature in °C.

c) Directions of magnetization of sandstone lens and adjacent lava in transition zone.
○ lava—south pole down ● sandstone—south pole down
Lower hemisphere stereogram.

Directions of magnetization of basal lavas and underlying baked sandstone at Sani Pass.
Lower hemisphere stereographic plot
× N pole down (lava) ▲ N pole down (baked sandstone).

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reversed lavas. If the transition represented simply a period during which the proportion of the magnetic minerals, or the degree of interaction between those minerals changed then one would expect to find only the two directions of magnetization opposite to one another. Complications which could produce the observed oblique directions are: (i) that at some subsequent time the rocks in the transition zone were partially magnetized in a direction different from that of the field during cooling and that the intensity of the original T.R.M. and/or this later magnetization varied across the transition; and (ii) that the magnetic minerals in the lavas of the transition zone have a preferred orientation and magnetic anisotropy similar to that of the Allard Lake haemo-ilmenites described by Hargraves (1959).

If in (i) the later magnetization were of isothermal remanent type one would expect it to have been removed by the a.c. demagnetization. The curves in Figure 2(b) and 4 show that this was not the case. On the other hand if the later magnetization were T.R.M. then it is necessary to postulate a magnetic component in the transition zone lavas which has a lower Curie point than the magnetic constituents of the reversed and normal lavas, for otherwise the lavas both above and below the transition would also have been remagnetized in the oblique direction. A series of thermal demagnetization experiments was carried out to ascertain, whether any such component existed. For this purpose 2 samples from the normal, 2 from the reversed and 6 from the transition zone were heated in a nitrogen atmosphere successively to temperatures of 100, 150 ... 600 °C in a non-inductively wound, non-magnetic tube furnace, and then cooled. A Helmholtz coil system was used to cancel the Earth's field. The residual field was less than 600γ. The directions of magnetization remained unchanged up to the Curie point of the magnetite. Figure 4(b) shows the directions of magnetization of a transition zone specimen from a height of 1230 ft above the base of the succession in the Maseru area. This specimen is typical. Since the directions of magnetization remain unchanged until the Curie point of the magnetite is reached, it follows that possibility (i) falls away.

Considering (ii) it may be remarked that no preferred grain orientation was observed in the polished sections. This finding is in accord with the observations of Stockley (1947, p. 53) that directional flow is rare in these lavas. Moreover, the conditions of eruption of the lavas were quiet and there is no sign of stress. Under these conditions it would not be expected that the ore minerals which represent a late stage in the order of crystallization would have formed under conditions promoting grain fabric. Finally, it would not be expected that even if preferred grain orientation had influenced the direction of magnetization, the preferred orientation would have varied in such a way as to produce the very diverse directions of magnetization shown in Figure 4.

5. The interbedded sandstone in the transition zone

A thin sandstone lens, approximately 3 ft thick, occurs at an elevation of 1100 ft above the base of the lavas in the Maseru area. Its lower contact with the lavas is sedimentary, whereas its upper edge has been baked by the overlying lava flow. The directions of magnetization of the baked sandstone and the overlying lava flow (Figure 4(c)) are very nearly the same, but the inclination is considerably less than that in the typical normal and reversed zones. It is difficult to explain this observation on the basis of a self-reversal mechanism for one would then have to

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assume preferred orientation and magnetic anisotropy of the ferromagnetic minerals, and further the preferred orientation and anisotropy would have to be the same in both the sediment and the lava. This must be regarded as improbable.

6. The Cave Sandstone

Six cores were drilled from the baked Cave Sandstone immediately below the base of the lavas. Five of the cores were collected at Sani Pass and the other in the Maseru area. The directions of magnetization of the 22 baked sandstone specimens from Sani Pass and of the 21 lava specimens immediately above the contact are plotted in Figure 4(d). The directions of both the lavas and the sandstones are reversed but the two directions do not coincide exactly. This is not an effect of misorientation for the cores were all drilled through the lava into the sandstones. The discrepancy cannot be explained in terms of a secondary soft magnetization for a.c. demagnetization showed no evidence of such a component. Moreover, the directions of magnetization remained unchanged up to 250 oersted and then commenced to scatter (see Figure 5). Specimens heated to 650 °C in a nitrogen atmosphere and allowed to cool in the Earth's field acquired a T.R.M. in the direction of the applied field. There is thus no inherent property of the rock which could account for the deviation. The deviation may be due to the change in secular variation during the time in which the magnetic constituents of the lava and those of the sandstone baked by the lava passed through their respective Curie points.

In the case of the baked sandstone from the Maseru area, the direction of magnetization is the same as that of the overlying lava (Figure 6(a)). The baked zone in the Maseru area is only 3 ft thick as against 7 ft at Sani Pass, and it is possible that the more rapid cooling implied by the thinner zone accounts for the coincident directions found at Maseru, whereas at Sani Pass the directions differed slightly.

Figure 6(a) shows that the direction of magnetization changes towards that of the present field as the distance from the contact increases. This is due to instability of the unbaked Cave Sandstone. Three samples of unbaked Cave Sandstone from the Sani Pass area were remeasured a few months after the initial series of measurements, and it was found that the directions of magnetization had changed significantly. A.C. demagnetization was attempted, but the intensities dropped too low for satisfactory measurement.

When examined under the ore microscope the magnetic material in the unbaked Cave Sandstone is seen to consist of well rounded grains of titano-magnetite. The titano-magnetite has been largely altered to maghemite and hematite in cracks and along the edges of the grains. The instability of the sandstones is presumably due to the presence of the maghemite (Akimoto & Kushiro 1960; Wilson 1961). In the baked sandstone there is some maghemite but the larger proportion has been converted to hematite. This probably accounts for the greater stability of the baked rock under a.c. demagnetization.

Samples of the baked and unbaked sandstones were heated to 650 °C in a nitrogen atmosphere, in air, and in a reducing atmosphere, and were then cooled in the Earth's field. In all cases the direction of magnetization was the same as that of the applied field. This observation serves to exclude all but the time-dependent self-reversal mechanism (v).

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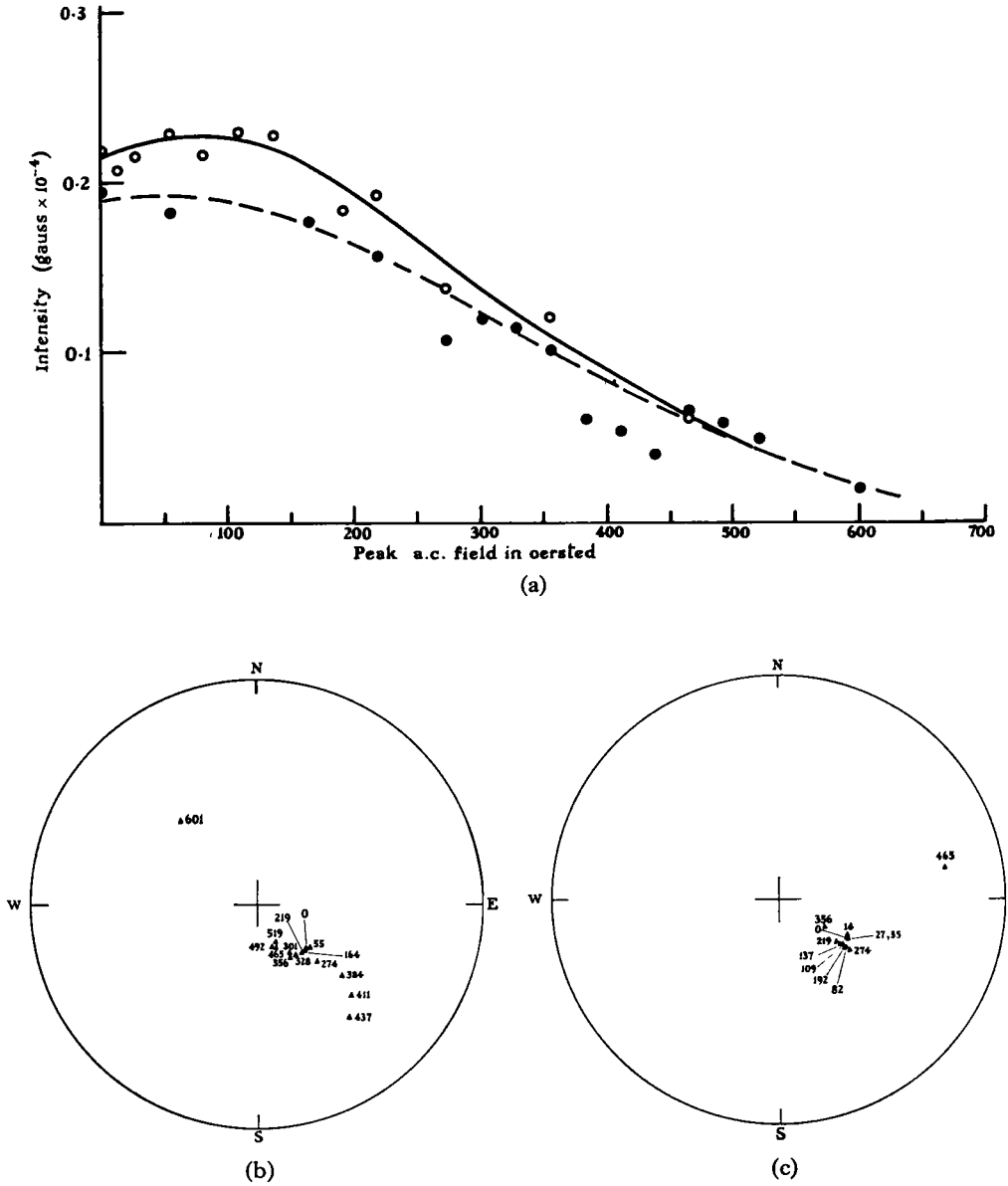


FIG. 5.—A.C. demagnetization curves for two baked sandstone samples underlying the lava at Sani Pass.

(a) Intensity of magnetization vs. peak a.c. field
—O— specimen 91/6
--●-- specimen 92/8

(b) (c) Lower hemisphere stereographic plots of changes of directions of magnetization with increasing a.c. field.
▲ N poles down

The numbers show peak a.c. field in oersteds.

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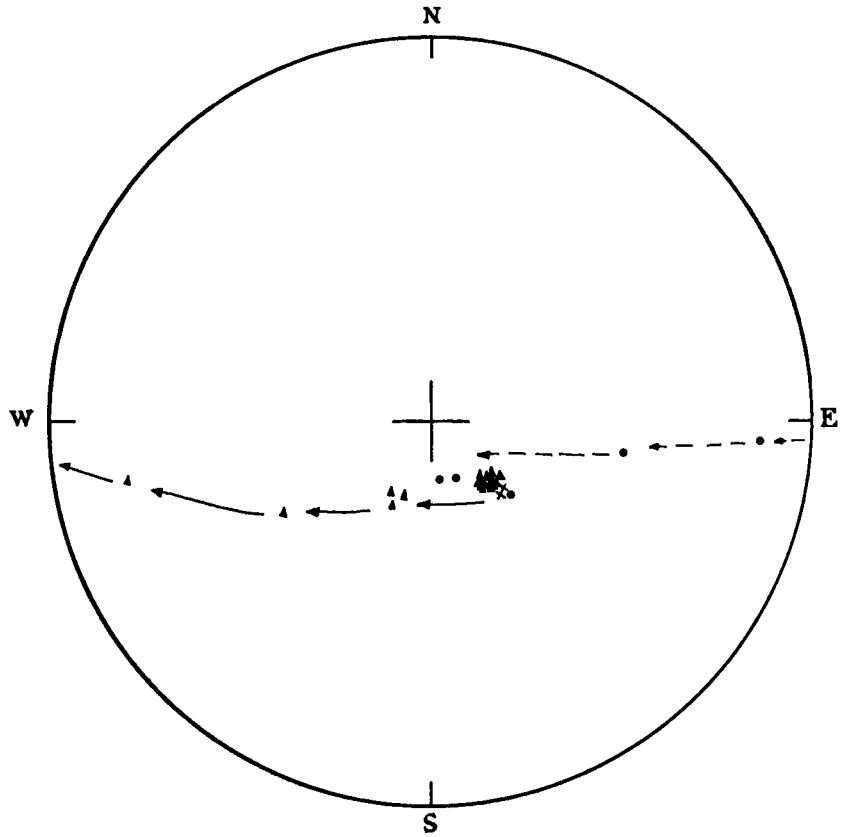


FIG. 6.—(a) Lower hemisphere stereographic plot of directions of magnetization of baked sandstone and the overlying lava in the Maseru area.

× N pole down (lava—6 specimens)
▲ N pole down (baked sandstone)
● S pole down (baked sandstone)
Increasing distance from lava contact indicated by arrows.

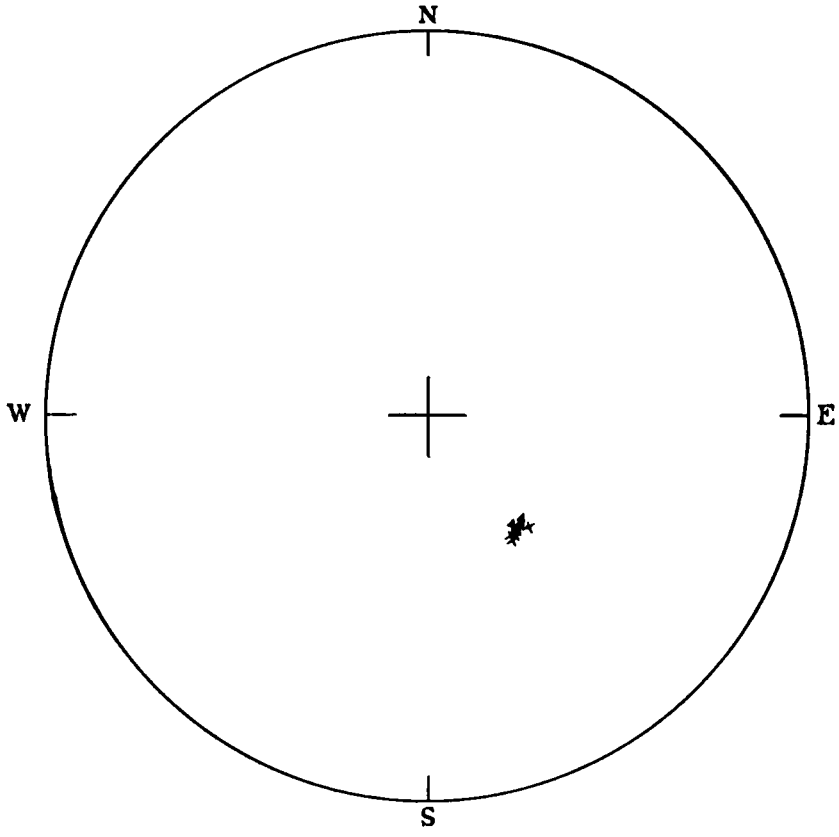


FIG. 6.—(b) Directions of magnetization of sandstone zenolith and adjacent lava.

Lower hemisphere stereographic plot
× lava N pole down. 5 specimens
▲ zenolith N pole down. 5 specimens

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7. The baked sandstone zenolith

A baked sandstone zenolith about 6 ft square occurs at a height of 180 ft above the basal lavas at Sani Pass. The direction of magnetization of the baked sandstone coincides with that of the adjacent lava (Figure 6(b)) and both were reversed. A.C. demagnetization established that the magnetization was stable. When heated and cooled in the Earth's field the direction of magnetization was that of the applied field.

8. The dykes cutting the lavas

At about 650 ft above the basal lavas in the Maseru area two normally magnetized dolerite dykes of thicknesses 6 and 10 ft were found cutting reversely magnetized lavas. The fact that the dolerite dykes were magnetized normally throughout their entire width and that the immediately adjacent reversed lavas are remagnetized in a normal direction coincident with those of the dykes suggests strongly that the original reversed direction of magnetization of the lava is to be ascribed to reversal of the Earth's field for the same flow is reversely magnetized further away from the dyke intrusion. A.C. demagnetization established the stability of magnetization of the normal dykes, the reheated lavas which were normally magnetized and the reversely magnetized lavas of the same flow further from the dykes. When specimens of the three rock types were heated and cooled in the Earth's field all acquired a magnetization in the direction of the applied field thus excluding all self-reversal mechanisms except the time-dependent ones. As the times of cooling of the lava flow when first extruded, and after reheating by the normal dyke would be of somewhat the same order and much greater than is possible in laboratory experiments, it seems difficult to account for the reversal of this lava flow on the basis of a time-dependent self-reversal mechanism.

Furthermore, the X-ray crystallographic analysis showed that the magnetic minerals in the reheated lava now normally magnetized had the same cell dimensions as the reversed lavas elsewhere in the sequence (see RbN in Table 1). It is of course possible to postulate that the reversed lavas acquired their magnetization by a self-reversal process, and that thereafter some process occurred as a result of which the lavas when reheated by the normal dyke no longer possessed the self-reversing property. Having regard to the similarity of properties of the reheated lava and that which was not reheated, shown by the X-ray analysis, the authors regard such a postulate as extremely artificial and improbable.

One normally magnetized dyke was found cutting normally magnetized lavas in the Maseru area. In this case the baked zone was also normal. Stability was de-established by a.c. demagnetization. No question of any self-reversal arises in this case.

General discussion

The authors consider that the evidence presented in this paper justifies the conclusion that there was a reversal of the Earth's field in Triassic–Jurassic times. It is, of course, true that some of the arguments are not conclusive in themselves, but the alternative hypotheses which would be required would be different for the different cases discussed and are individually improbable, so that when all lines of evidence are considered together it seems very probable that the main magnetic field changed from reverse to normal during the period of extrusion of the lavas.

Wilson (1961) has come to a similar conclusion from a study of the Tertiary lavas of Northern Ireland.

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In reviewing the geophysical aspects of rock magnetism Runcorn (1955) remarked that:

"If, by and large, observed reversals are due to a reversal of the geomagnetic field, one would expect the following:

"(1) In the interval between opposed directions of magnetization there should be in some instances zones of transitional directions of magnetization. Such transitional zones could hardly arise from a Néel mechanism.

"(2) Over the whole of geological time during which it is apparent that hundreds of reversals have occurred, the geomagnetic field should not be biased towards one polarity relative to the axis of rotation.

"(3) During times in which the polarity remains constant over a sufficient time to establish geologically the contemporaneity of different strata concordant directions of magnetization should be found in strata of the same age. Apparently the results from Palaeozoic rocks in Great Britain indicate that this test may be feasible.

"(4) No systematic differences in the physical or chemical properties of the iron oxide minerals in oppositely magnetized rocks should be found."

It is thought that the work reported in this paper is in accord with what Runcorn suggested should be expected on the basis of a reversal of the geomagnetic field.

It may be remarked that the fact that normal and reverse directions of magnetization were found in the lavas, the directions being close to those found by Graham & Hales (1957) for the dolerite intrusives, suggests that the dolerites and lavas were contemporaneous rather than that the dolerites were the late hyperbyssal phase as has been suggested by Walker & Poldevaart (1949).

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